

KARNATAKA RESIDENTIAL EDUCATIONAL INSTITUTIONS SOCIETY

ICT Class 9 - Formative Assessment 1

Formative Assessment (FA-1)

Class: Class 9

Assessment Type: Formative Assessment (FA-1)

Total Marks: 30

Duration: 90 minutes

Topics Covered: Python Fundamentals - Review and Advanced Concepts

General Instructions

1. All questions are compulsory.
2. Read each question carefully before answering.
3. Write neatly and legibly.
4. Marks are indicated against each question.

Part A: Written Concept Paper (20 Marks)

Section I: MCQ (5 x 1 = 5 marks)

a. What is the output of the following Python code?

```
x = [1, 2, 3, 4, 5]
print(x[1::2])
```

- a. [1, 3, 5]
- b. [2, 4]
- c. [1, 2, 3]
- d. Error

b. Which of the following is a valid variable name in Python?

- c. 2nd_place
- d. _score
- e. total-marks
- f. class

c. What will be the output?

```
print(type(10 // 3))
```

- a. <class 'float'>
- b. <class 'int'>
- c. <class 'decimal'>
- d. Error

d. Which operator is used for string repetition in Python?

- e. +
- f. *
- g. **
- h. %

e. What is the output of `bool("")`?

- f. True
- g. False

- h. None
- i. Error

Section II: Short Answer (5 x 2 = 10 marks)

- a. Differentiate between `is` and `==` operators in Python with examples.
- b. Write the output of the following and explain:

```
a, b = 5, 10  
a, b = b, a  
print(a, b)
```
- c. Explain the difference between `while` loop and `for` loop. When would you use each?
- d. Write a Python program to check if a given number is a palindrome without converting it to a string.
- e. What is the purpose of `break`, `continue`, and `pass` statements? Give one example of each.

Section III: Long Answer (5 x 1 = 5 marks)

- a. Write a Python program that takes a list of integers and returns a new list containing only the prime numbers from the original list. Include proper function definition, docstring, and test the function with at least 3 different lists.

Part B: Hands-on Practical Lab Task (5 marks)

Write a Python program that:

- Takes a sentence as input from the user
- Counts the number of words, vowels, consonants, digits, and special characters
- Displays the frequency of each vowel
- Checks if the sentence is a palindrome (ignoring spaces and case)
- Uses functions for each counting operation

Part C: Notebook Maintenance (5 marks)

- a. Neatness and organisation of notebook (2 marks)
- b. Completion of all assigned classwork and homework (2 marks)
- c. Proper headings, dates, and indexing (1 mark)